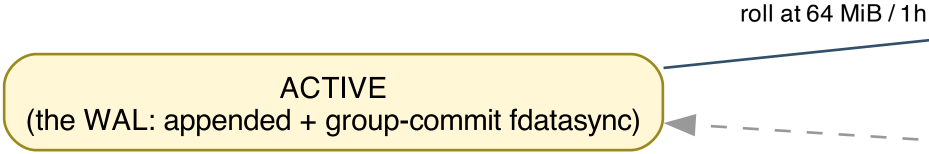
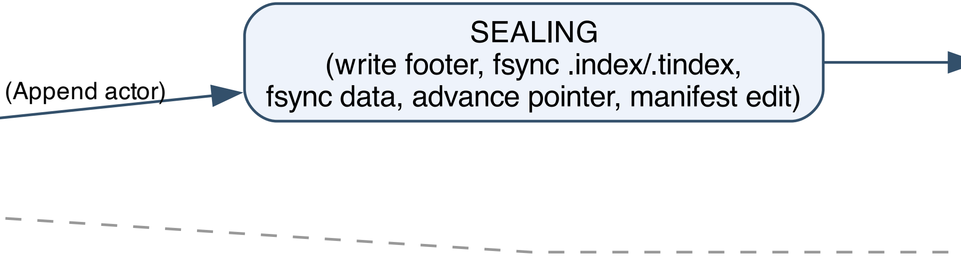
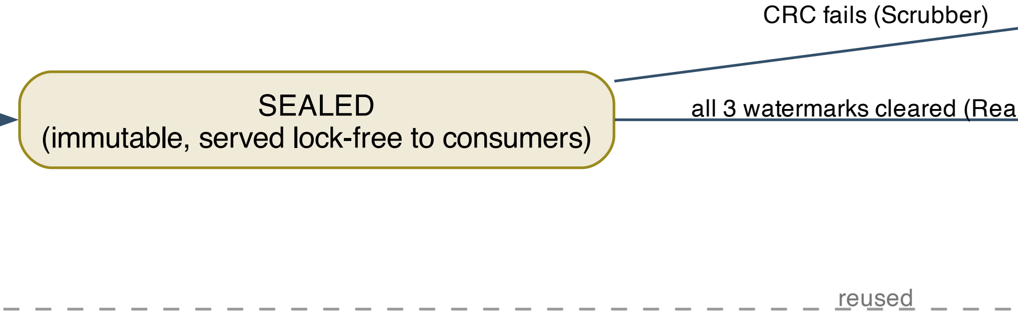
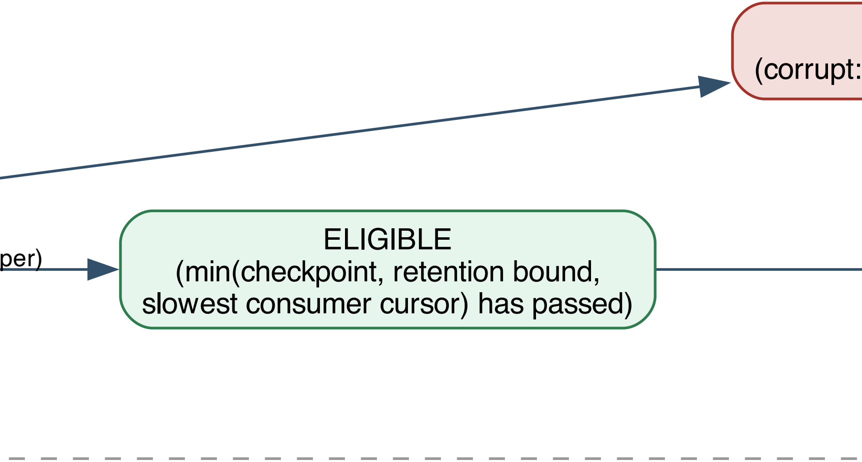


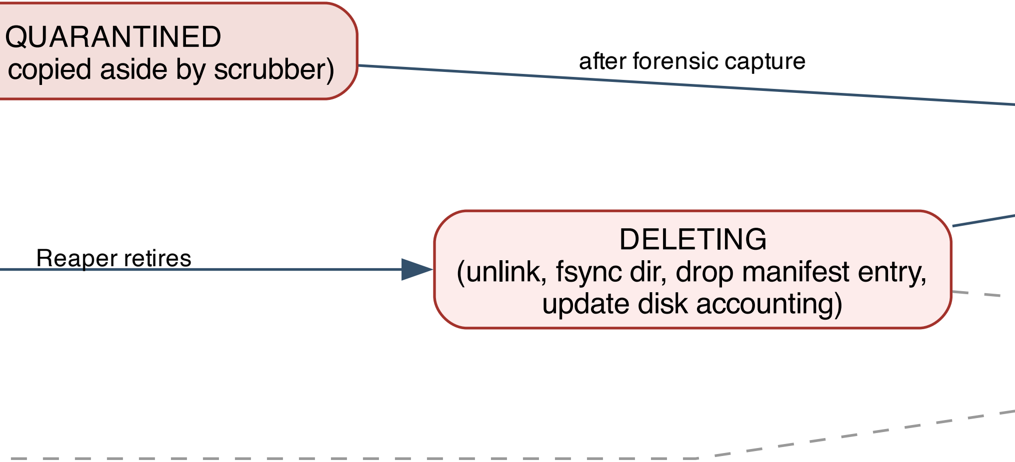




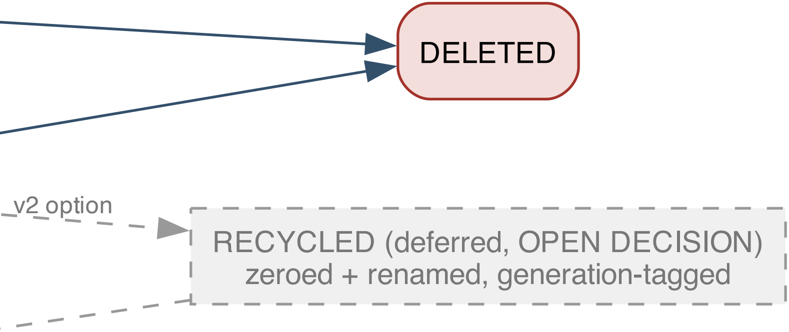
IronBus segment lifecycle an



d file cleanup (log-is-WAL): who retires the files,



and when



Cleanup ownership (no single actor is blocked by another):
Append actor (#4): seals the active segment, hands immutable index to readers.
Group-commit thread (#6): establishes durable_offset; fatal fsync error freezes the insta
Checkpointer (#7): publishes the durable watermark (gate 1 for deletion) on roll / 1s / 4
Reaper / janitor (#13): retires sealed segments past min(checkpoint, retention, slowest
event-driven on seal, 60s age timer, forced run at 90% disk high-water.
Compactor (#13, optional): re-copies survivor segments, never in-place, single atomic
Backpressure controller (#10): when nothing is eligible and disk fills, durable spills the
(drop_new, counted), telemetry drop_oldest, block is opt-in and bounded

ance.
MiB.
cursor);

manifest swap.
n sheds
.

Lessons borrowed: PostgreSQL checkpoints + WAL recycling,
Kafka whole-segment retention (active never deleted), SQLite

RocksDB max_total_wal_size + flush-advances-log, wal_autocheckpoint, etcd snapshot-then-compact.